Worksheet 1-10 A Types of Discontinuities #1

AP Calculus Name_____

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Directions: Find and classify any x value(s) where the functions below have a discontinuity.

1. $f(x) = \dfrac{x^2 - 4}{x^2 - 5x + 6}$

2. $g(x) = \dfrac{x^3 - 2x^2 - 3x}{x^2 - 3x - 4}$

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3. Let k be the function defined by $k(x) = \dfrac{x^4 - 4x^2}{x^2 - 4x}$. Which of the following statements is true?

A) k has a discontinuity due to a vertical asymptote at $x = 0$ and at $x = 4$.
B) k has a removable discontinuity at $x = 0$ and a jump discontinuity at $x = 4$.
C) k has a removable discontinuity at $x = 0$ and a discontinuity due to a vertical asymptote at $x = 4$.
D) k is continuous at $x = 0$ and has a discontinuity due to a vertical asymptote at at $x = 4$.

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4. Let g be the function defined by $g(x) = \frac{x^3 - 9x}{x^2 - x - 6}$. Which of the following statements about g at $x = -2$ and $x = 3$ is true?

A) g has a jump discontinuity at $x = -2$, and g is continuous at $x = 3$.
B) g has a jump discontinuity at $x = -2$ and a removable discontinuity at $x = 3$.
C) g has a discontinuity due to a vertical asymptote at $x = -2$, and g is continuous at $x = 3$.
D) g has a discontinuity due to a vertical asymptote at $x = -2$, and g has a removable discontinuity at $x = 3$

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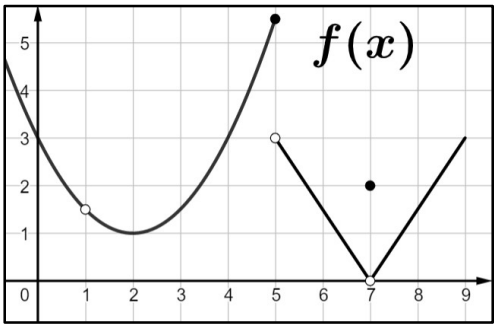
5. The graph of $f(x)$ is shown below. What are all the values of x for which f has a removable discontinuity?

A) 1 only B) 5 only C) 1 and 7 only D) 1, 5, and 7



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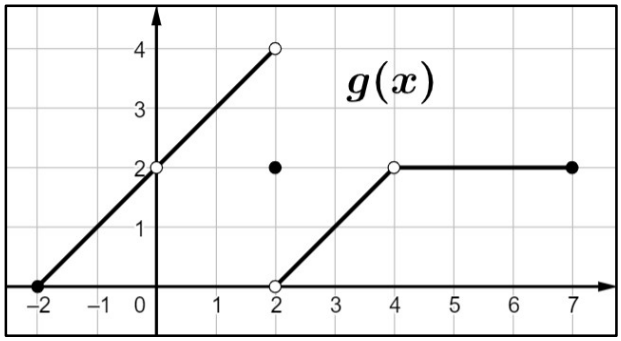
6. The graph of $g(x)$ is shown at the right. For which value(s) of x does g have a removable discontinuity?

A) 0 only B) 2 only C) 0 and 4 only D) 0, 2, and 4



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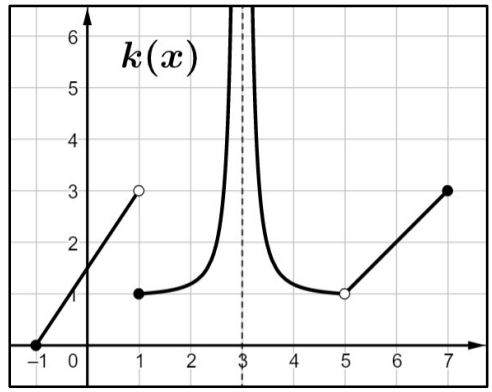
7. The graph of $k(x)$ is shown at the right. Let h be the function defined by

$$h(x) = \frac{x^2 - 6x + 5}{x^2 - 4x + 3}$$

For which value(s) of x do the functions h and k **both** have a nonremovable discontinuity?



$$f(x) = \begin{cases} x^2 + 2, & x < 1 \\ \sin\left(\frac{\pi x}{2}\right) + 2x^2, & x \geq 1 \end{cases}$$

8. Let f be the piecewise defined function shown above. Find and classify any x values where $f(x)$ is discontinuous.

$$g(x) = \begin{cases} 5 - 4x, & x < 2 \\ -x^2 - 1, & x \geq 2 \end{cases}$$

9. Let g be the piecewise defined function shown above. Find and classify any x values where $g(x)$ is discontinuous.

$$h(x) = \begin{cases} 3x - 1, & x < 0 \\ \ln(x + 1) - e^x, & x > 0 \end{cases}$$

10. Let h be the piecewise defined function shown above. Find and classify any x values where $h(x)$ is discontinuous.

$$k(x) = \begin{cases} \dfrac{x^2 - 1}{x^2 - 3x - 4}, & x < 3 \\ 5 - 2x, & x \geq 3 \end{cases}$$

11. Let k be the piecewise defined function shown above. Find and classify any x values where $k(x)$ is discontinuous.

$$f(x) = \begin{cases} c^2 - x, & x < 3 \\ cx + 7, & x > 3 \end{cases}$$

12. Let f be the piecewise defined function shown above and let c be a positive constant. Find the value of c such that $f(x)$ has a hole at $x = 3$.